

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2021 - 22 Examination

Semester: 6**Subject Code: 203105395****Subject Name: Software Testing & Quality Assurance****Date: 07-04-2022****Time: 10:30 am to 01:00 pm****Total Marks: 60****Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions**(15)**

1. CMM stands for _____.
2. Data Flow Testing is a type of _____ testing.
3. _____ is called module under test.
(a) Driver (b) Stub (c) Software (d) None of the above
4. A methodology for quality improvement of customer satisfaction and benefits is
(a) SPICE (b) TQM (c) Six Sigma (d) CMM
5. _____ requirement refers to companies involved in design, production, manufacture and maintenance of products
(a) ISO 9001 (b) ISO 9002 (c) ISO 9003 (d) All of the above
6. What is not the factor of software quality?
(a) Reliability (b) Portability (c) Functionality (d) Efficiency
7. Which type of testing will be done at User site?
8. _____ are the quantitative measures used to estimate the progress, quality, productivity and health of the software testing process.
9. While an application is running, suddenly restart the computer, and afterwards check the validness of the application's data integrity. It is an example of _____ testing.
10. Cyclomatic Complexity is computed as _____.
(a) $V(G) = \text{The number of regions}$
(b) $V(G) = E - N + 2$ where E is the number of flow graph edges, N is the number of flow graph nodes.
(c) $V(G) = P + 1$ where P is the number of predicate nodes contained in the flow graph G
(d) All of the above
11. Boundary value analysis is a _____ type of testing Technique.
12. Risk should include:
(a) Probability of occurrence a problem (b) Impact of problem (c) Test cases (d) Both a&b
13. _____ testing is also called as monkey testing.
14. ECP stands for _____.
15. _____ is also known as dynamic testing, where we are ensuring that "we have developed the product right."

Q.2 Answer the following questions. (Attempt any three)**(15)**

- A) Explain Boundary value analysis technique with example.
- B) What is decision-based table testing? Draw decision table for login screen?
- C) Explain different identifiers of Test plan?
- D) Differentiate verification and validation.

Q.3 A) Draw and explain McCall's Quality Factor in detail.**(07)****B) Explain black box testing techniques in detail.****(08)****OR****B) Explain fault based and scenario-based testing in detail.****(08)**

Q.4 A) What are the primary objective when we test a software. Briefly outline the requirement specification of an ATM. Write the test cases on ATM transactions? **(07)**

OR

A) Explain Basis Path testing with example. **(07)**

B) Explain testing in multiplatform environment in detail. **(08)**